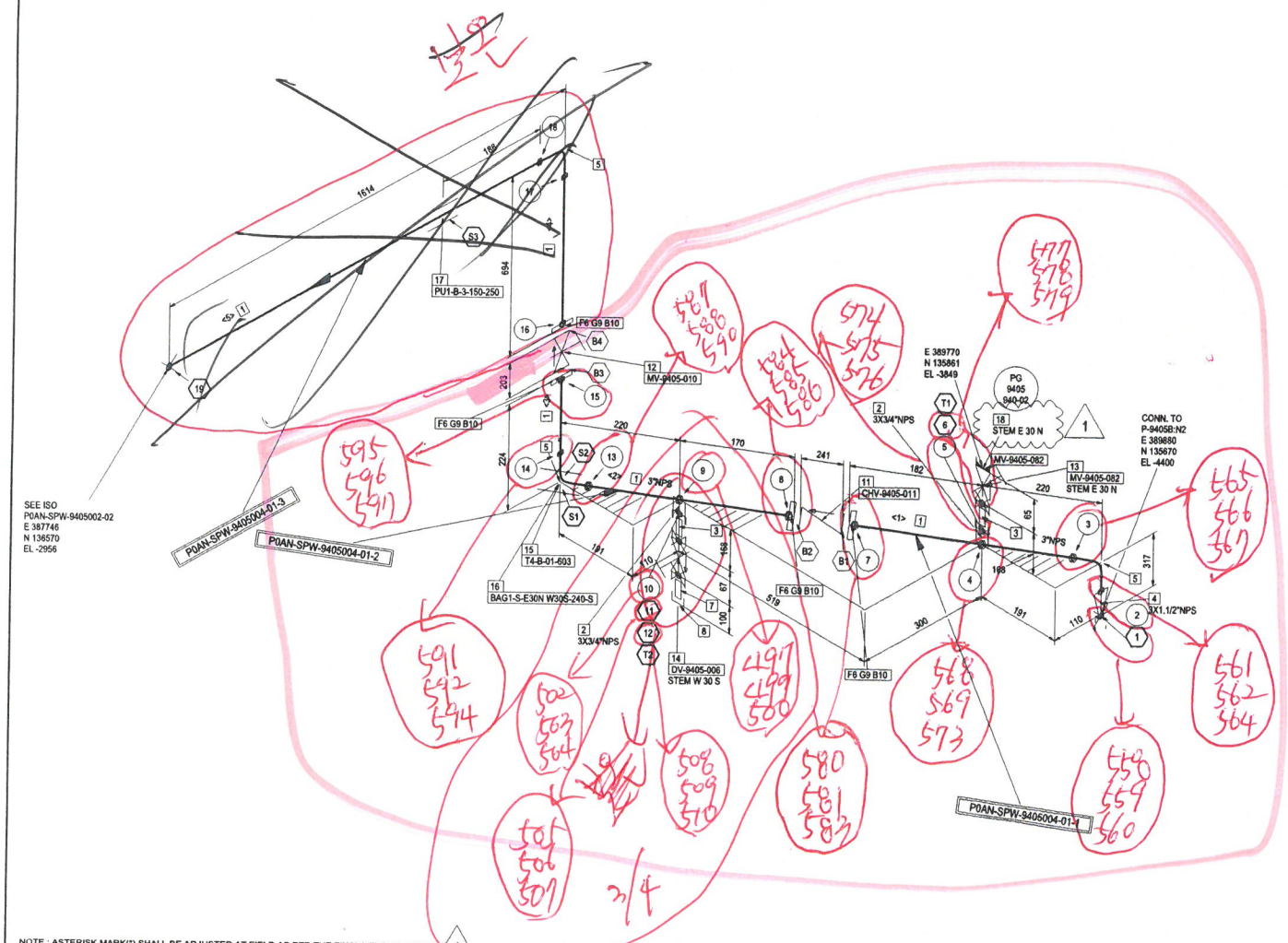
K1-JHC-940-SPW1-007



NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (℃)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (℃)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
0.75	S98	2.7	25 ~ 80	5	128	-	S	ISN	25	-	N	5	K1-103
1.5	S98	2.7	25 ~ 80	5	126	-	S	ISN	25	-	N	5	K1-103
3	S98	2.7	25 ~ 80	5	126	-	S	ISN	25	-	N	5	K1-103

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
1	14-06-25	ISSUED FOR REVISION	H.S.JUNG	M.H.SUNG	T.H.LIM
0	15-04-25	ISSUED FOR INFORMATION	H.S.JUNG	M.H.SUNG	T.H.LIM

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BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	3"	2.8 M	S98	P	PA02-000001903
2	HALF COUPLING SW CL3000 ASTM A182-F316/316L	3X3/4"	2	S98	TH	PB99-000038951
3	PIPE NIPPLE PBE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	2	S98	NPPG	PB99-000044539
4	CON. SWAGE BLE/PSE SMLS ASTM A403-WP316/316L SCH10S * SCH40S	3X1.1/2"	1	S98	RCS	PB99-000102550
5	90 DEG. ELBOW (LR) BW WELDED ASTM A403-WP316/316L SCH10S	3"	3	S98	E	PB99-000038410
6	SLIP-ON FLANGE RF CL150 ASTM A182-F316/316L	3"	4	S98	F	PB01-000023956

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
7	PIPE NIPPLE TOE(NPT)/POE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPTG	PB99-000123907
8	CAP SCR'D (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
9	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHIT E 4.5MM(0.18") THK.	3"	4	S98	G	PF01-000008268
10	100 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UNIU NR x MM	5/8"	16	S98	B	PB99-000076890
11	CHECK VALVE FLGD-RF CL150 ASTM A351-CF8M SS316 TRIM BC SWING DISC (VA C161)	3"	1	S98	VC	PP06-000002611
12	GATE VALVE FLGD-RF CL150 ASTM A351-CF8M SS316 TRIM STELLITE SEATS BB OS&Y FLEX. WDG DISC HW. OP (VAV151)	3"	1	S98	VA	PP02-000010434
13	GATE VALVE SW*SCR'D (NPT) CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS&Y SOLID WDG DISC HW. OP (VAV005A)	3/4"	1	S98	VAST	PP02-000010451
14	GATE VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS &Y SOLID WDG DISC HW. OP (VAV005)	3/4"	1	S98	VA	PP02-000010443
15	T4-B-01-603	3"	1			T4-B
16	BAG1-S-E30N W30S-240-S	3"	1			BAG-S
17	PU1-B-3-150-250	3"	1			PU1-B

INSTRUMENTS

18	TAG NO : PG-9405-940-02, INSTRUMENT	3/4"	1			INSTRUMENT

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	278	3		<2>	266	3	
<3>	100	3		<4>	570	3	
<5>	1500	3					

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Project PROVIDENCE

Approval For Construction
PIPING MANAGER Bea Chang Won
2025-06-14

PROJECT
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LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	P0AN
P&ID NO.	K1-DSL-UT-PID-9405			PROJ. NO.	16480	FLUID	SPW
GAD NO.	-			LINE NO.	3-SPW-9405004-S98-ISN		
DWG NO.	P0AN-SPW-9405004-01			SH'T NO.	1 OF 1	REV.	1

PIPE modified at 17:03 4 Jun 2025, Drawing Created at 17:04:56 4 June 2025
PIPE NAME : 3-SPW-9405004-S98-ISN_P0AN-01